

Weekly Summary #6 | Association Rule Mining

Part 3 | Which cell type sits next to which

GVHD Analysis | Week of: September 23, 2026

Muscle issue: how much muscularis a biopsy caught differs by stage, and in opposite directions in the two organs - it rises with stage in the duodenum and falls in the colon. Any result involving Muscle therefore mixes biology with what the biopsy happened to include, so every figure below that touches it is shown twice: all FOVs, then only those that sampled the muscularis. The restricted version drops a few patients from each severe group, so it is a check on the others rather than a replacement.

Rules of a cell type with itself: a rule may name the same cell type twice — *Muscle beside Muscle*. These are **kept everywhere in this note**, and each one lands on the diagonal of its grid. Every subtitle says whether they are in or out. Section 4 is about what they show; Section 2 is the one place where keeping them changes what a reader should conclude, so it draws that figure both ways.

Counting a rule fairly: a rule is counted only in an FOV that could have carried it - one holding enough of both of its cell types to mine. Every fraction under a dot panel is out of those FOVs, and every ring on a grid is the same thing drawn as a circle.

1 Claim 1 - the direction of a rule is anatomical

$A \rightarrow B$ and $B \rightarrow A$ are different measurements here, and most mined rules fire one way round only. This is not an artefact of the thresholds: a rule and its mirror have genuinely different lift, because a neighbourhood is centred on a cell, so *epithelium beside a Paneth cell* and *a Paneth cell beside epithelium* count different things.

In one image, two squares are filled and their mirrors are empty. Paneth cells have epithelium beside them; smooth muscle has endothelium beside it. Neither reverse rule was mined — which is what the anatomy predicts, since Paneth cells are a fraction of a percent of the tissue and almost no epithelial cell has one nearby.

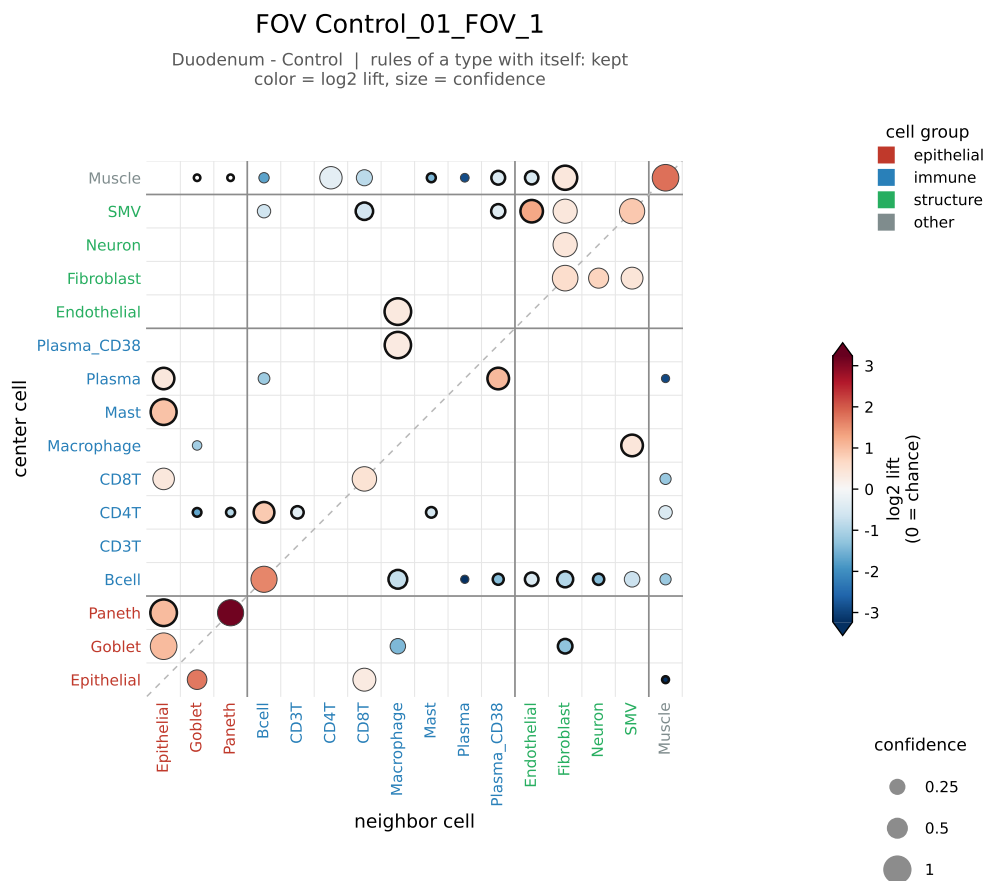


Figure 1: Reading the grid: each row is a cell type taken as the centre of a neighbourhood; each column is a cell type found within 25 μm of it. A dot means that rule was mined, and a heavy black outline means the mirror rule was not — it fired one way round only. A dot on the dashed diagonal is a cell type beside its own kind; it is its own mirror, so it is never outlined. Colour is log2 lift, so twice as likely and half as likely sit equally far from white. Dot size is confidence. Here: one healthy duodenal FOV, axes trimmed to the cell types it uses.

Across the cohort the asymmetry holds. Paneth $\rightarrow$ Epithelial fires in most of the control FOVs that could carry it and its mirror in none of them; SMV $\rightarrow$ Endothelial sits above chance at every stage while its mirror is rare. The panel below shows both pairs beside their mirrors, so the gap can be read directly.

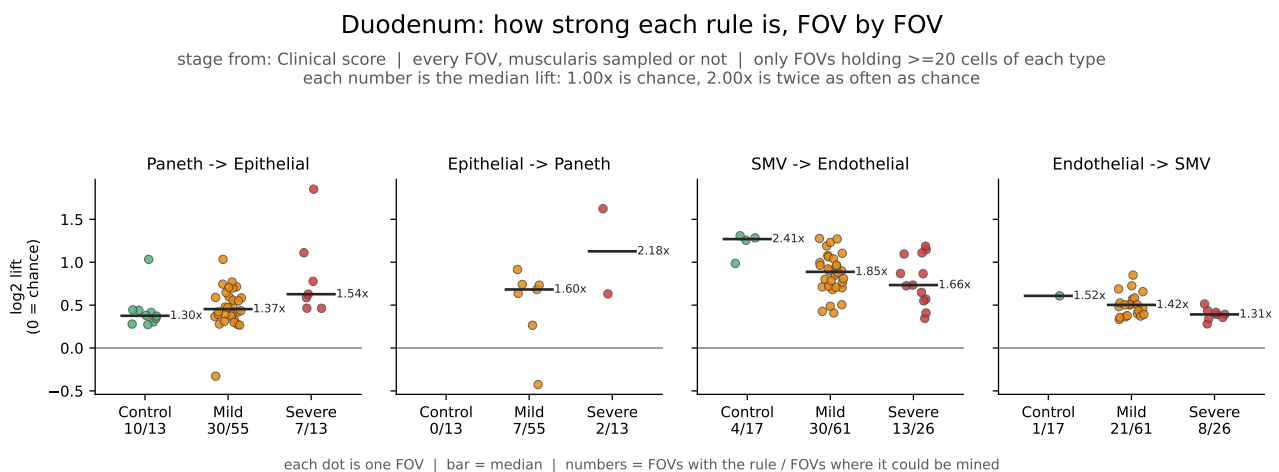


Figure 2: Reading a dot panel: one cell pair per panel; each dot is one FOV carrying the rule, the bar is the median, and the number beside it is the median lift (1.00x is chance). The fraction under each stage is FOVs with the rule out of the FOVs where it could have been mined. The two right panels are the mirrors of the two left.

Pooled, the grid is visibly not symmetric about its diagonal, with the asymmetry the right way round.

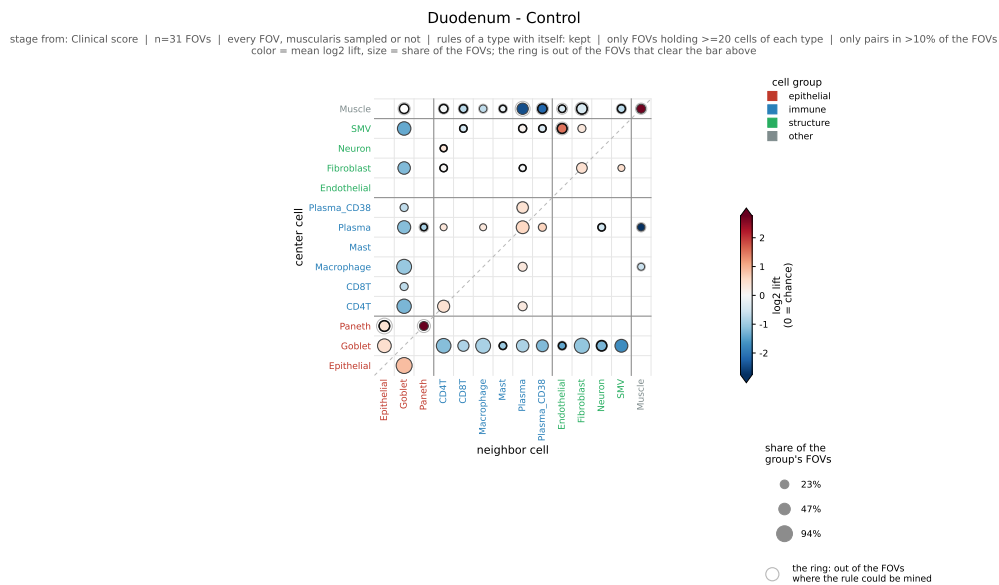


Figure 3: Healthy duodenum, all its FOVs pooled, axes trimmed as before.

2 Claim 2 - the rules reconstruct the layers of the gut

The gut is built in layers — epithelium, lamina propria, muscularis — and **nothing about this was given to the mining**. No coordinates, no tissue regions, no anatomy: only which cell types were found within $25\mu\text{m}$ of each other. If the rules put that structure back, they are measuring the tissue rather than the cell-type table.

Folded into lineages, the grid shows the layering. The diagonal is warm — a layer sits with itself — and the epithelial row is cold against everything outside the epithelium. Epithelium against muscle is the strongest avoidance anywhere in the data.

which part of the tissue sits next to which

stage from: Clinical score | every FOV, muscularis sampled or not | rules of a type with itself: kept | only FOVs holding ≥ 20 cells of each type
 color = mean log2 lift, size = mean share of the pairs in the box

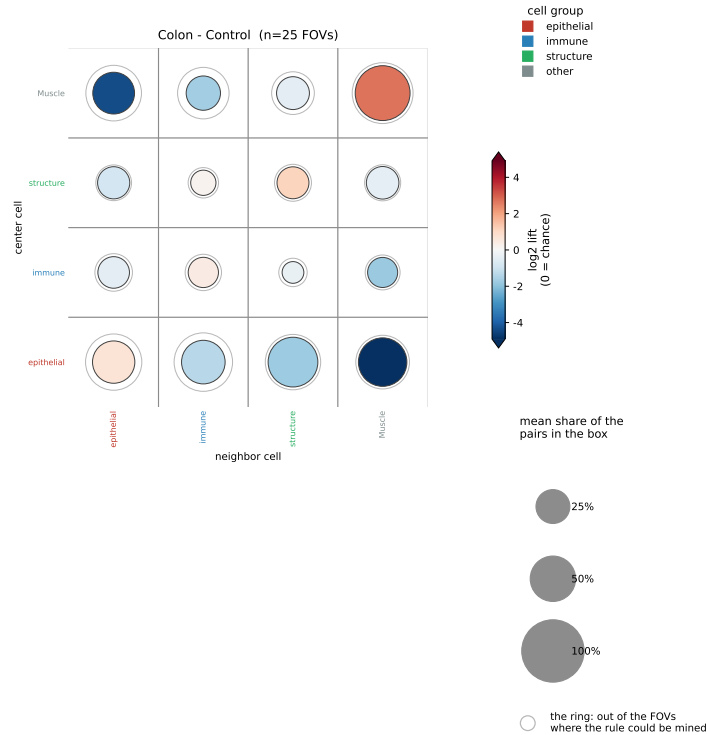


Figure 4: Healthy colon, the cell types folded into their lineages; Muscle and Brunner glands keep their own row and column, forming a lineage with nothing else. A box averages the log2 lift of every rule inside it, **weighted by how many FOVs each rule appears in** — otherwise a rule seen in one FOV would count as much as one seen in nearly all of them.

The same picture repeats in both organs and at every stage, and this is the stereotypic organisation Azulay *et al.* describe for healthy gut, recovered from rules alone. That it barely moves with disease is the second half of the result rather than a disappointment: a method that invented structure would drift, and this one does not, so the changes it does report are worth taking seriously. Claim 3 is where they are.

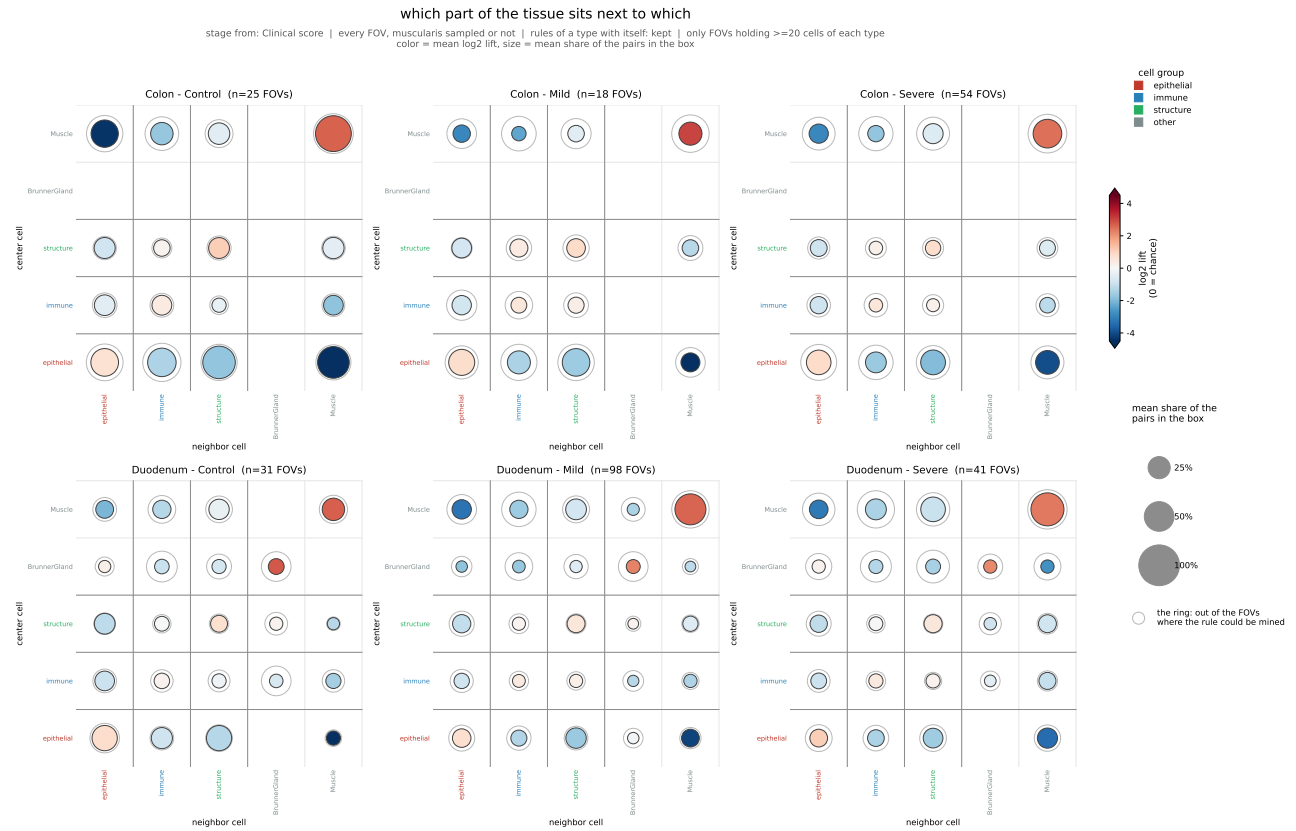


Figure 5: All six organ $\times$ stage groups, folded as in Fig. 4.

The warm diagonal is a layer sitting with itself, not only a cell type sitting with itself. A rule naming one cell type twice lands on the diagonal, so a warm *epithelial* box could in principle be nothing but Goblet beside Goblet. Dropping every such rule cools the diagonal a little and leaves every box off it untouched, since no such rule ever lands there. The layering is the stronger of the two statements.

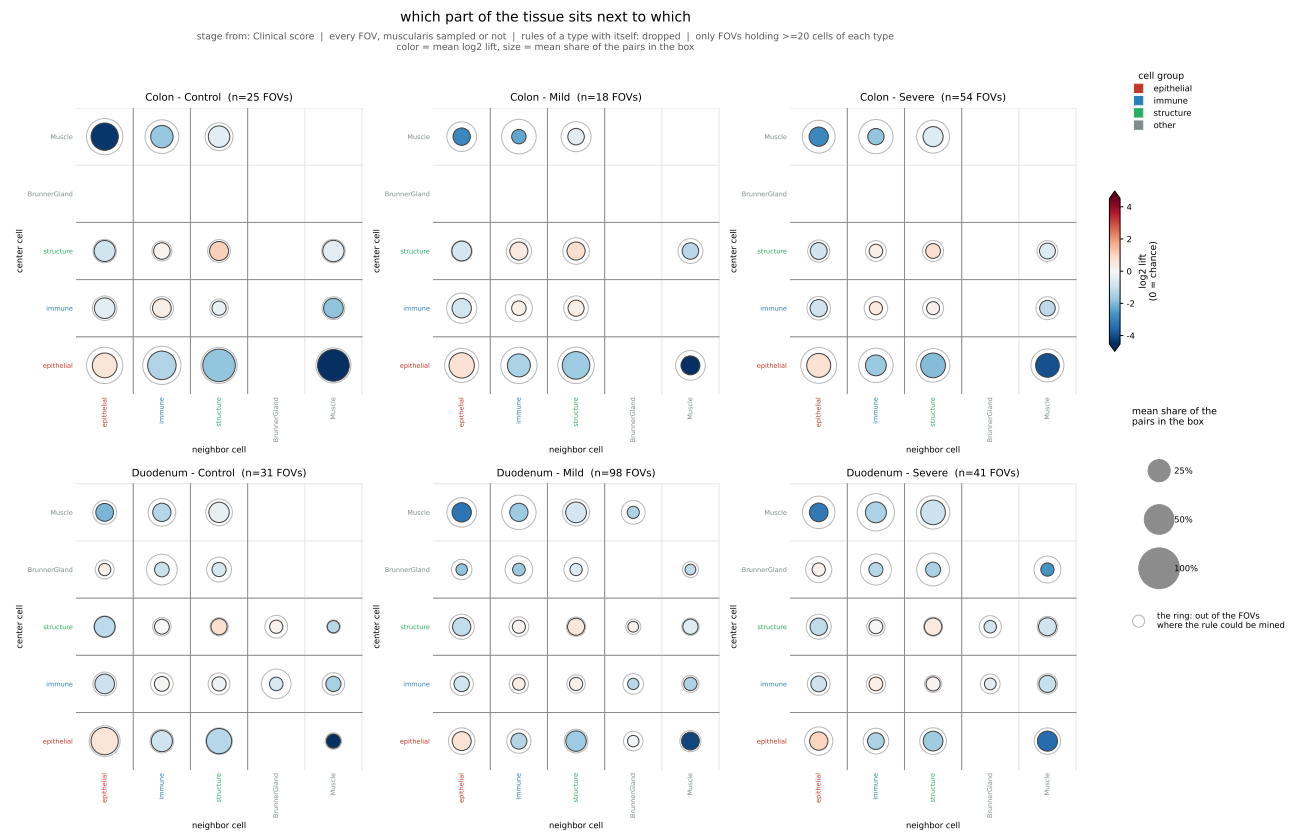
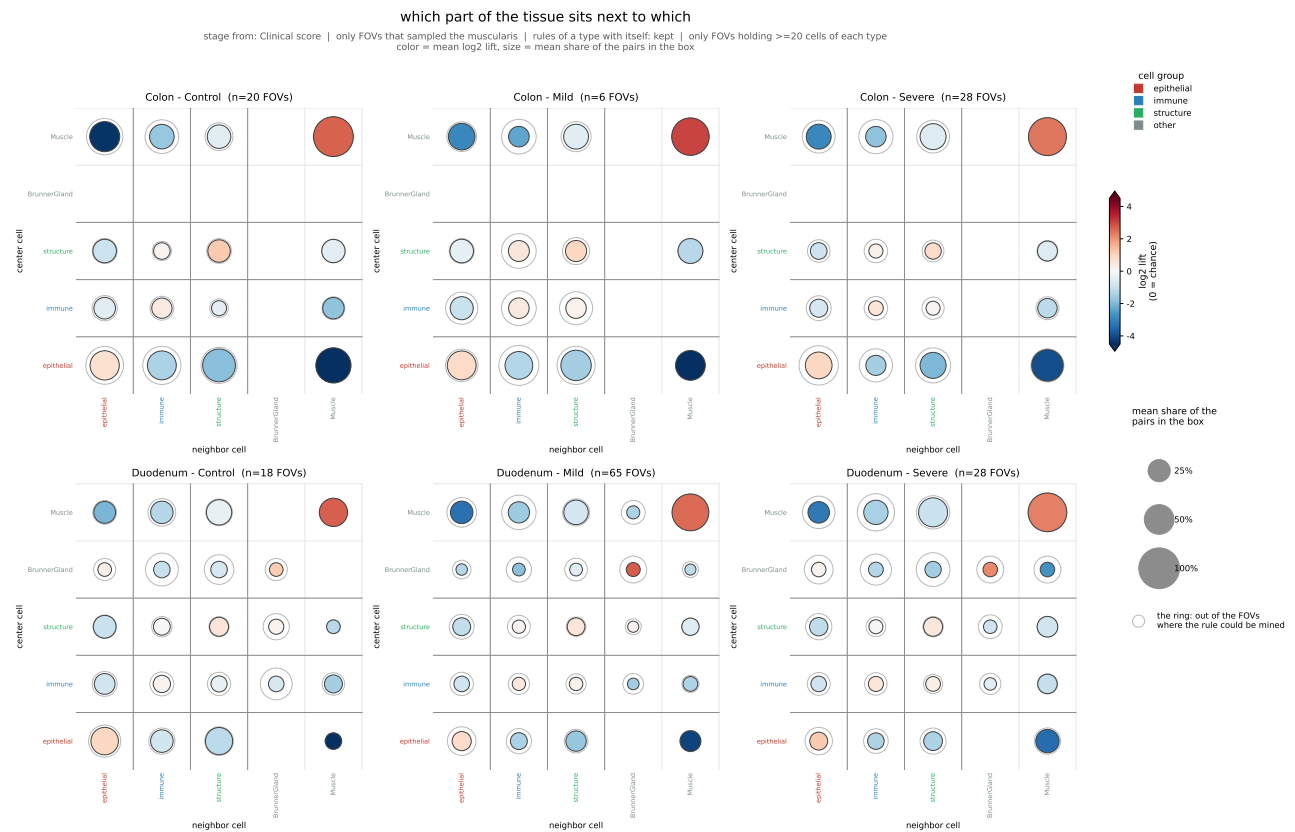


Figure 6: Fig. 5 with every rule of a cell type with itself dropped. The Muscle and Brunner gland boxes on the diagonal empty out, because those two lineages hold one cell type each and had nothing but their own rule in them.

Dropping the FOVs with no muscularis changes nothing but the Muscle row. The diagonal, the epithelial row and the stromal block are the same picture, so the layering is not an artefact of which biopsies caught muscle.



Unfolded, each pair sits at the same height in every stage — the boundary does not loosen. Tested one rule at a time, with a Mann-Whitney on log2 lift and Benjamini-Hochberg correction, almost no across-layer rule moves toward chance between control and severe. Several tighten instead. The paper's "less organised" is about villus and crypt *shape*; a rule only asks which cell types end up within $25\mu\text{m}$ of each other, and a crypt can lose its shape while its cells keep their neighbours.

Colon: how strong each rule is, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

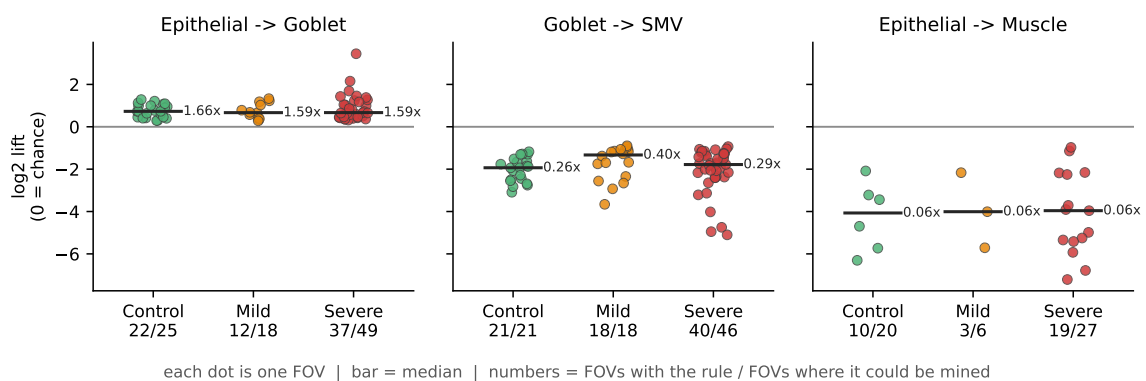


Figure 8: Colon, all FOVs. Three pairs across the layer boundary, drawn as in Fig. 2.

Epithelial $\rightarrow$ Muscle is the one pair the sampling could have invented, and it survives. On muscularis FOVs only it stays far below chance at every stage; the other two pairs are unaffected, as they involve no muscle.

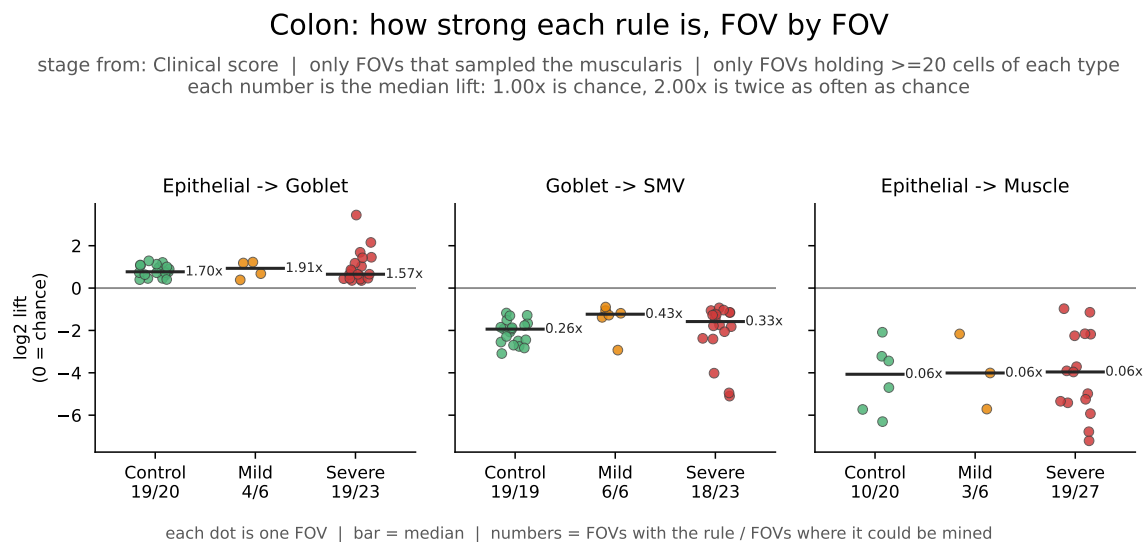


Figure 9: Fig. 8 restricted to FOVs that sampled the muscularis.

3 Claim 3 - a rule is not its two cell types, counted

Everything so far could still be answered with a cell-type table: if a rule only fired where its two cell types happened to be common, the method would be an expensive way of counting. Lift is built not to be — it divides by how common each cell type is, so a rule away from chance is already a claim about arrangement. This section tests that directly, and it is the claim the method stands on.

How strong a rule is has nothing to do with how common its cells are. In healthy tissue the two are uncorrelated (Spearman $\rho = 0.03$): a rule between two abundant cell types is no stronger than one between two rare ones. Whatever the grid is measuring, it is not the cell-type table read twice.

And the two can move in opposite directions. A handful of rules change strength between control and severe *against* the direction both of their cell types went — the cells go one way and the arrangement goes the other. They are named in red on the right of the figure: Paneth and Goblet against epithelium and Plasma.CD38 against Plasma all *tighten* while both their cell types thin out, and SMV beside SMV *loosens* while those cells become more common.

is a rule only its two cell types, counted?

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type | pairs with ≥ 5 FOVs in control and in severe

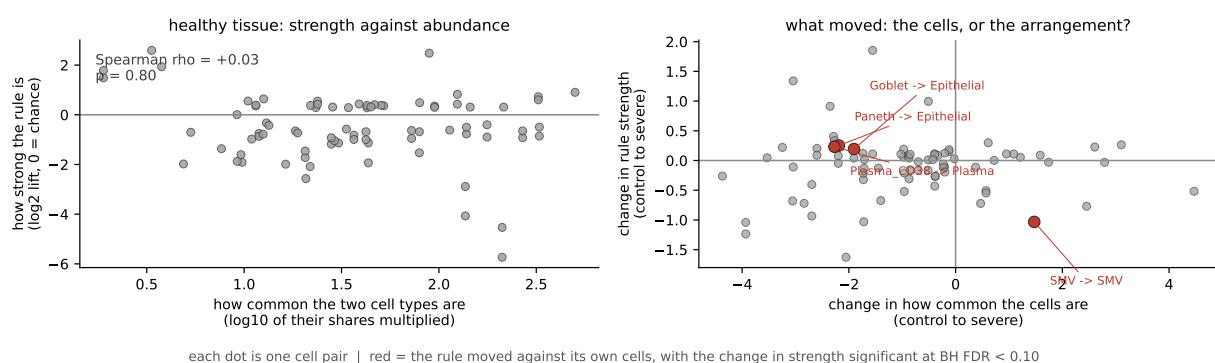


Figure 10: Left: each dot is a cell pair in healthy tissue — how common its two cell types are, against how strong the rule between them is. The cloud is flat. **Right:** how far each moved between control and severe. A dot in the upper left or lower right moved against its own cells; those are drawn in red and named.

This is the part counting cannot reach. A cell-type table says *Paneth cells are disappearing*. The rule says *the Paneth cells still there are packed against epithelium more tightly than they were in health* — a statement about what the surviving tissue looks like, not about how much of it is left. Only the second needs spatial data.

One diseased image already shows the composition changing too. Against Fig. 1, the Paneth row has gone and an Endocrine row has appeared.

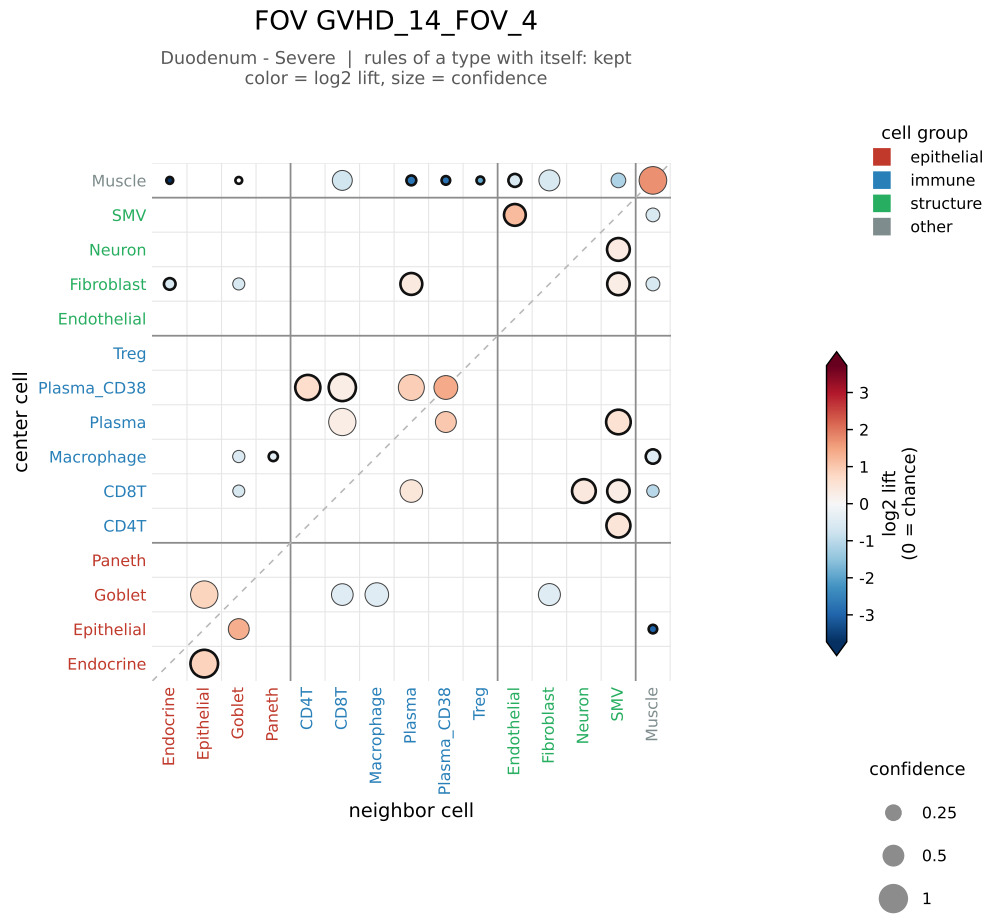


Figure 11: One severe duodenal FOV, drawn as Fig. 1.

The paper's compositional findings are visible here too, and they are the weaker half. Paneth → Epithelial thins out with disease; Endocrine → Epithelial cannot be measured in healthy duodenum at all, because no control FOV holds enough endocrine cells, and appears once disease sets in. These track the Paneth loss and endocrine gain the paper reports. They confirm the rule table against a known result; they do not add to it, and on their own they would not be worth the machinery.

Duodenum: how strong each rule is, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

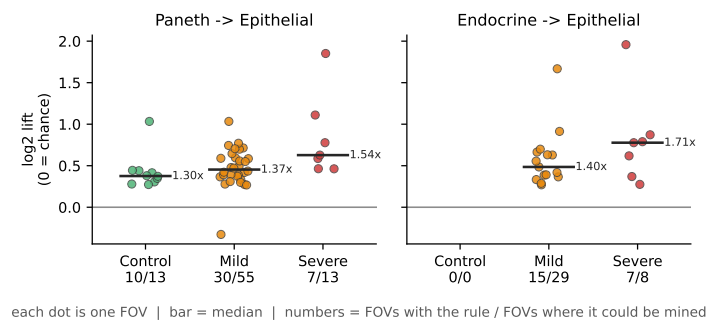


Figure 12: Duodenum, drawn as in Fig. 2. The height of a dot is the rule's strength; the count beneath is how many FOVs carry it out of how many could.

Duodenum: how common each cell type is, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not
every FOV of the group counts here, whether or not any rule was mined in it

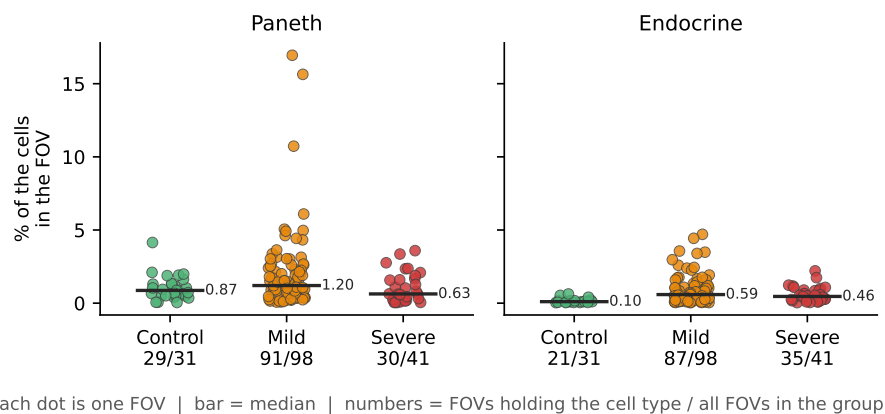


Figure 13: The same FOVs, now showing how much of each one the cell type makes up. Same layout as Fig. 12, so the two can be read against each other: the counts move, and Fig. 10 is what is left once that movement is divided out.

The Paneth rule is sharper against the pathological score than the clinical one, falling further in severe disease when the groups are cut by pathology. The endocrine rule rises either way. The Lerner pathological score is the histological readout while the Glucksberg clinical score grades symptoms — the paper notes the two disagree in six of its patients — so a rule tracking the tissue score more closely is a small point in the method's favour.

Duodenum: how strong each rule is, FOV by FOV

stage from: Pathological score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

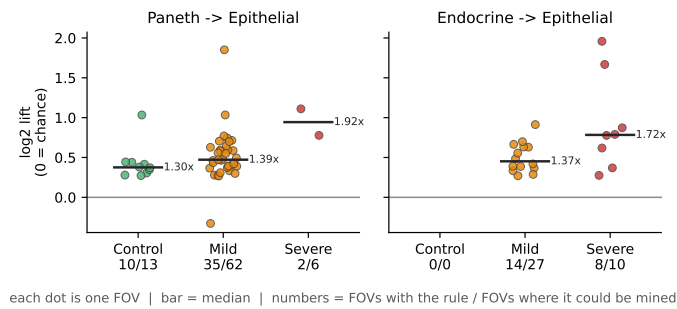


Figure 14: Fig. 12 regrouped by the pathological score. Everything else in this note uses the clinical score, as the paper does.

The colon repeats the pattern in both directions. Neutrophil $\rightarrow$ Epithelial is absent from control and mild disease and appears only in severe, though in too few FOVs to lean on; CD4T $\rightarrow$ Plasma is the firmer of the two and runs the other way, common in healthy colon and rare in severe.

Colon: how strong each rule is, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

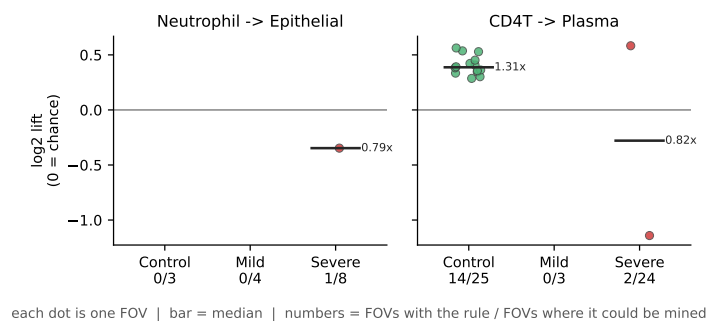


Figure 15: Colon, drawn as in Fig. 2.

Colon: how common each cell type is, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not
every FOV of the group counts here, whether or not any rule was mined in it

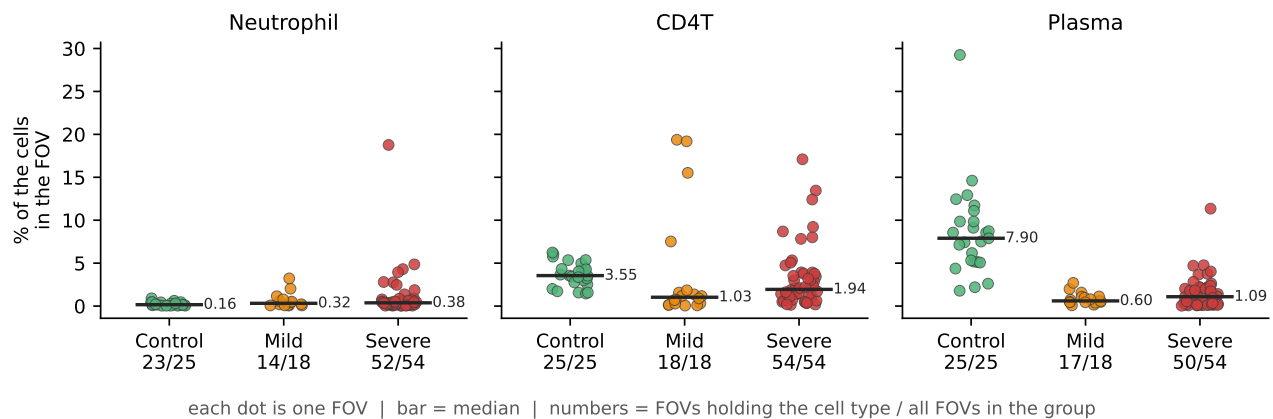


Figure 16: Colon cell abundance, drawn as in Fig. 13: neutrophils spread with severity while plasma and CD4 T cells fall away, which is the enrichment and depletion the paper reports.

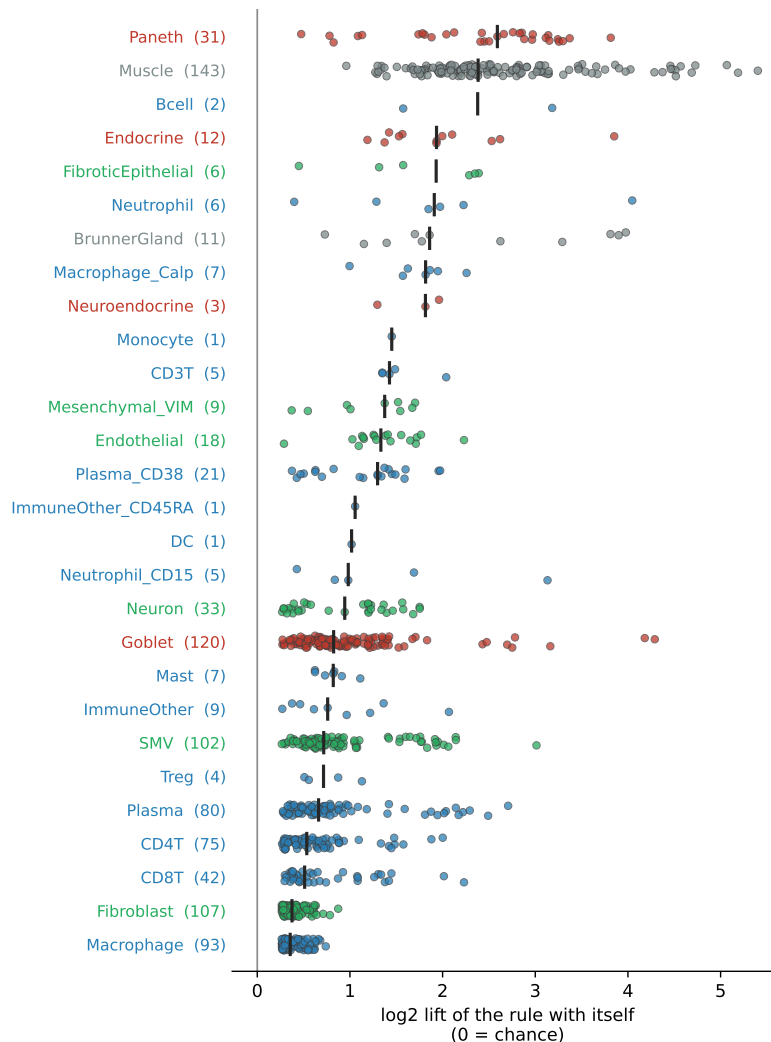
4 Claim 4 - a cell type sits with its own kind

A fair share of the mined rules name the same cell type twice, and for some of those the answer is known before the mining runs: muscle is a continuous band, and Paneth cells sit packed at the base of a crypt. Both should sit with their own kind, strongly. Both do, which makes this the cheapest check in the note — a method that got it wrong would be broken.

Not one cell type avoids its own kind. Every cell type carrying such a rule sits above chance. The strongest are the ones the anatomy names — Paneth, muscle, endocrine cells, fibrotic epithelium, Brunner glands — and the weakest are the ones spread thinly through the lamina propria, macrophages and fibroblasts and T cells, which is where cells that fill the tissue evenly belong.

how strongly a cell type sits with its own kind

every FOV carrying a rule of a cell type with itself | both organs, every stage



each dot is one FOV | bar = median | the number after a name is how many FOVs carry the rule

Figure 17: Every FOV carrying a rule of a cell type with itself, both organs and every stage pooled, drawn as in Fig. 2. The number after a name is how many FOVs carry the rule, and the names are coloured by cell group as on the grids above.

Three cell types carry no such rule anywhere, and one of them is Epithelial — the most common cell type in the data. That is the abundance handicap of the next section seen from the inside: a cell type in nearly every neighbourhood cannot clear a lift threshold, not even beside itself. The other two, APC and NK, sit at the opposite end, among the rarest.

Two of these rules follow the disease, and two make good controls. In the colon, Plasma beside Plasma is common in healthy tissue and rare in severe, which is the plasma cell loss the paper reports; in the duodenum, Paneth beside Paneth thins the same way. The controls stay put: Goblet beside Goblet holds throughout the colon, and Muscle beside Muscle through the duodenum, which is what a band of muscle should do whether or not the patient is ill. As in Claim 3, the count is what carries the signal — the heights barely move, except for Plasma, which climbs: the plasma cells still there sit closer together than they did in health.

Colon: how strongly a cell type sits with its own kind, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

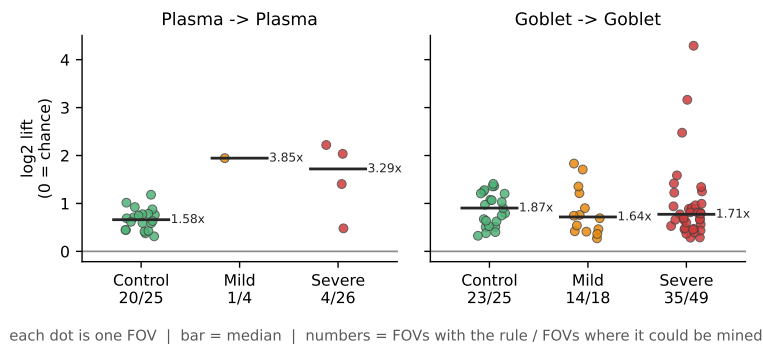


Figure 18: Colon, drawn as in Fig. 2.

Duodenum: how strongly a cell type sits with its own kind, FOV by FOV

stage from: Clinical score | every FOV, muscularis sampled or not | only FOVs holding ≥ 20 cells of each type
each number is the median lift: 1.00x is chance, 2.00x is twice as often as chance

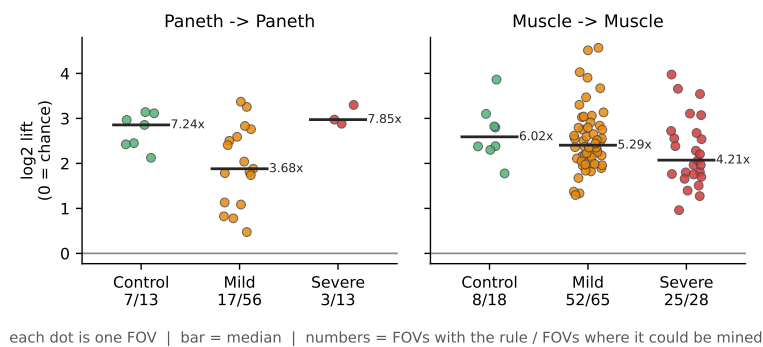


Figure 19: Duodenum, drawn as in Fig. 2.

5 All six groups at once

The Goblet rows say it most clearly: small dots inside large rings, so the rules hold wherever goblet cells remain and there are simply fewer such FOVs.

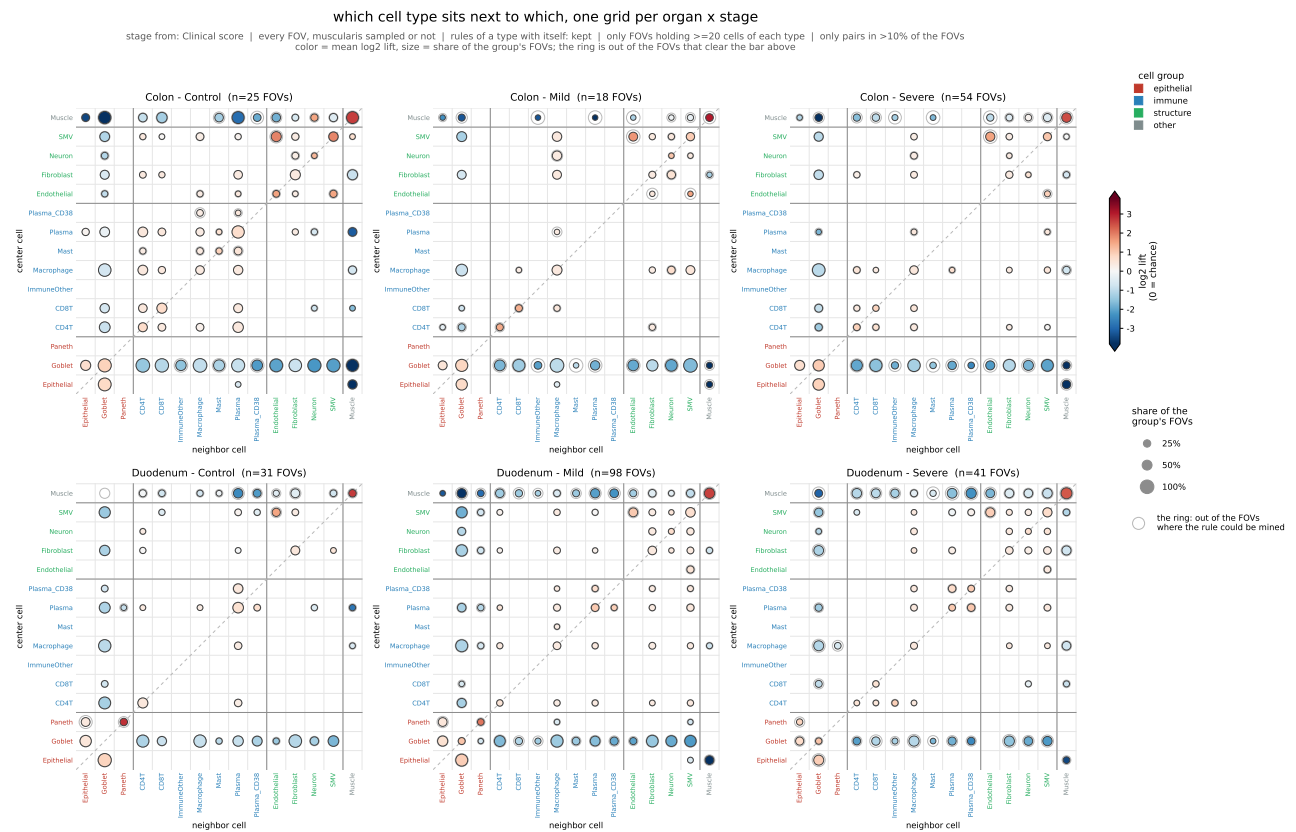


Figure 20: All six groups on a shared axis, restricted to the cell types carrying a rule in over a quarter of the FOVs of at least three groups.

On muscularis FOVs only, the epithelial changes are unchanged and the Muscle rows move. The Paneth, Endocrine and Goblet rows read the same, so the findings above do not depend on which biopsies caught muscle.

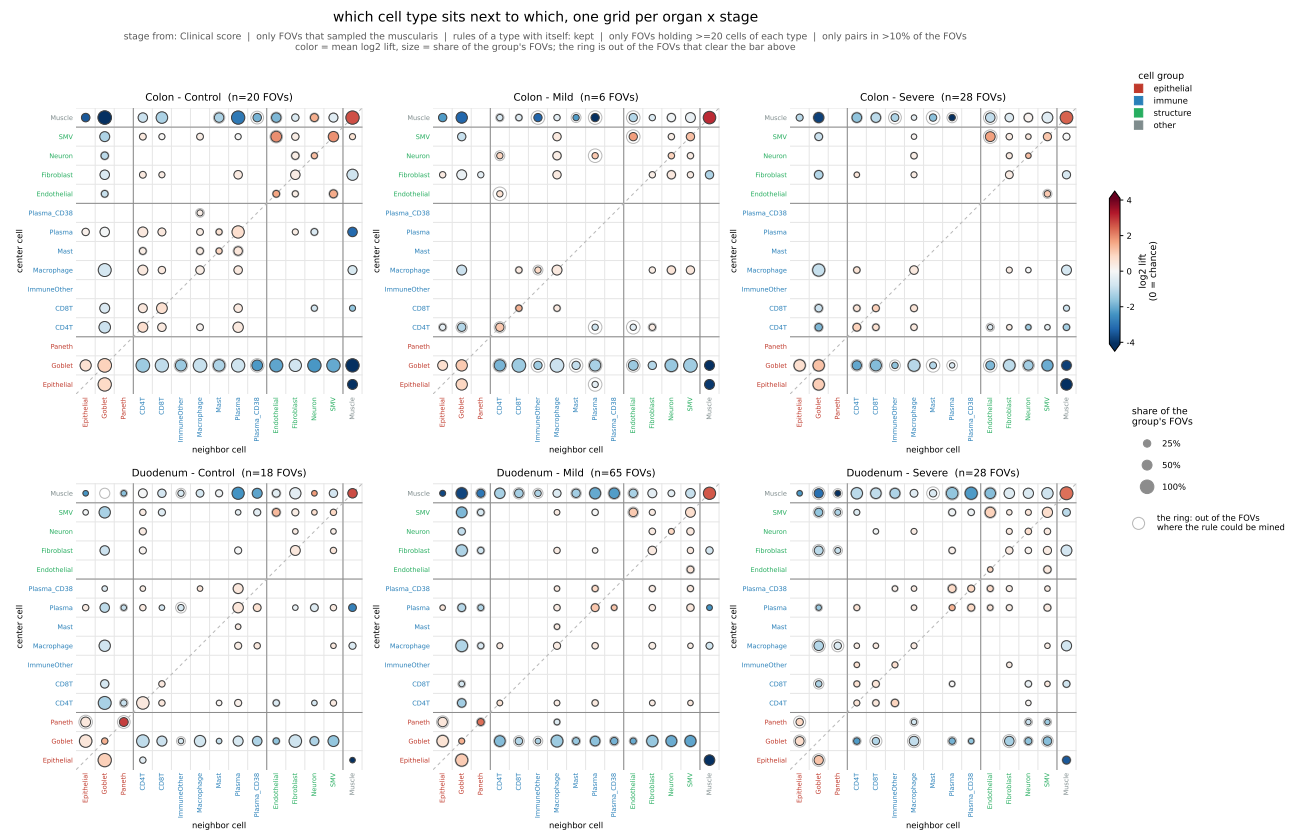


Figure 21: Fig. 20 restricted to FOVs that sampled the muscularis.

6 What the method cannot answer

The grid is sparse. A median FOV yields a few dozen rules out of nearly a thousand possible pairs, so an untrimmed grid is almost entirely empty. Trimming to the cell types actually used makes one grid far easier to read; the shared axis is kept only where grids must be compared.

Abundant cell types are handicapped. A cell type in nearly every neighbourhood has a lift near chance by construction and cannot clear the threshold. Epithelial cells are the case in point: the most common type in the data, and not one rule with themselves anywhere in the cohort. This also puts the paper's intraepithelial-lymphocyte result out of reach — the signs are right at every stage, $CD8T \rightarrow Epithelial$ above chance and $CD4T \rightarrow Epithelial$ below it, but the rules clear threshold in too few FOVs to test.

One FOV is coarser than one crypt. The paper's local-microenvironment result (their Fig. 6) is measured in a ring around each crypt, where an infiltrated crypt can sit beside an untouched one. A rule is mined over a whole FOV, which averages the two together. Within-FOV patchiness would need mining at crypt level — the natural next step.